

RFD900A User Manual

(English Translation)



RFD900A Product Specifications and Performance

RFD900A Configuration

RFD900A Modem Tools User Manual

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1. Main Features

The RFD900A is a compact, high-performance wireless data transmission module. Its main features include:

- Non-directional transmit/receive design. Each module integrates a USB-to-serial chip. USB and UART interfaces switch automatically, with USB taking priority.
- Operating frequency range: 902–928 MHz, ISM band
- Transmit circuit uses a low-noise amplifier design greater than 20 dB, with added low-pass filter unit
- Adjustable transmit power, maximum 1W transmit power
- Receive circuit features high-sensitivity LNA and SAW front-end
- Outdoor line-of-sight communication distance up to 40 km or more (depending on antenna performance and communication rate requirements)
- Maximum air data rate up to 250 Kbps
- Dual antenna interfaces, supports antenna diversity and differential antenna configuration
- Supports local radio AT command configuration and remote radio RT command configuration
- Module temperature monitoring function; automatically adjusts duty cycle based on temperature to avoid overheating
- Operating temperature: -40 to +85°C
- Dimensions: 33 mm × 53 mm × 9.5 mm
- Weight: 23 g

2. Feature Parameters

Performance

Parameter	Value
Communication Rate	4, 8, 16, 19, 24, 32, 48, 64, 96, 128, 192 and 250 kbps
Indoor Distance	500 m – 1 km
Outdoor Line-of-Sight Distance	40 km or more, depending on antenna performance
Transmit Power	0 to 30 dBm, adjustable in 1 dBm steps
Receive Sensitivity	>-121 dBm at low rates, >TBA at high rates
Low Noise Amplifier	>20 dB

Features

Parameter	Value / Description
Serial Data Interface Level	Standard 3.3 V, maximum withstand 5 V
Configuration Method	AT commands, APM Planner, custom configuration tool
Operating Frequency Band	902 MHz – 928 MHz
Spread Spectrum Technology	FHSS (Frequency Hopping Spread Spectrum)
Serial Port Rate	2400, 4800, 9600, 19200, 38400, 57600, 115200 bps
Optional Antennas	Yagi, ½ wave dipole, ¼ wave monopole
Compliance Standards	FCC Part 15.247, AS/NZS 4268:2008

Network Security

Parameter	Value / Description
Addressing Method	Network ID: 0 – 499
Channels	Up to 50 frequency hopping channels
Topology Support	Point-to-point, Point-to-multipoint (requires 2.X firmware for multipoint)

Power Options

Parameter	Value
Supply Voltage	Rated 5 V, min 4 V, max 5.5 V
Transmit Current	~1 A (maximum power mode)

Receive Current	~60 mA
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3. Voltage, Current and Power Specifications

Figure 3.1: The graph below shows the relationship between input voltage and output power.

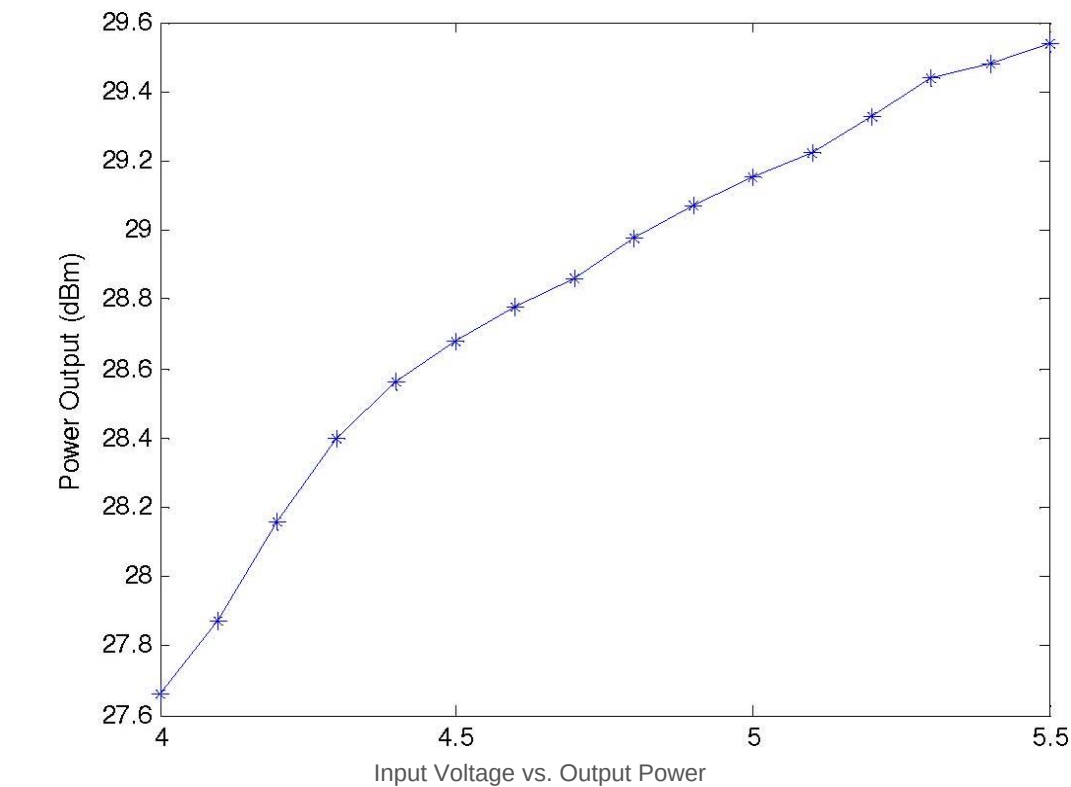
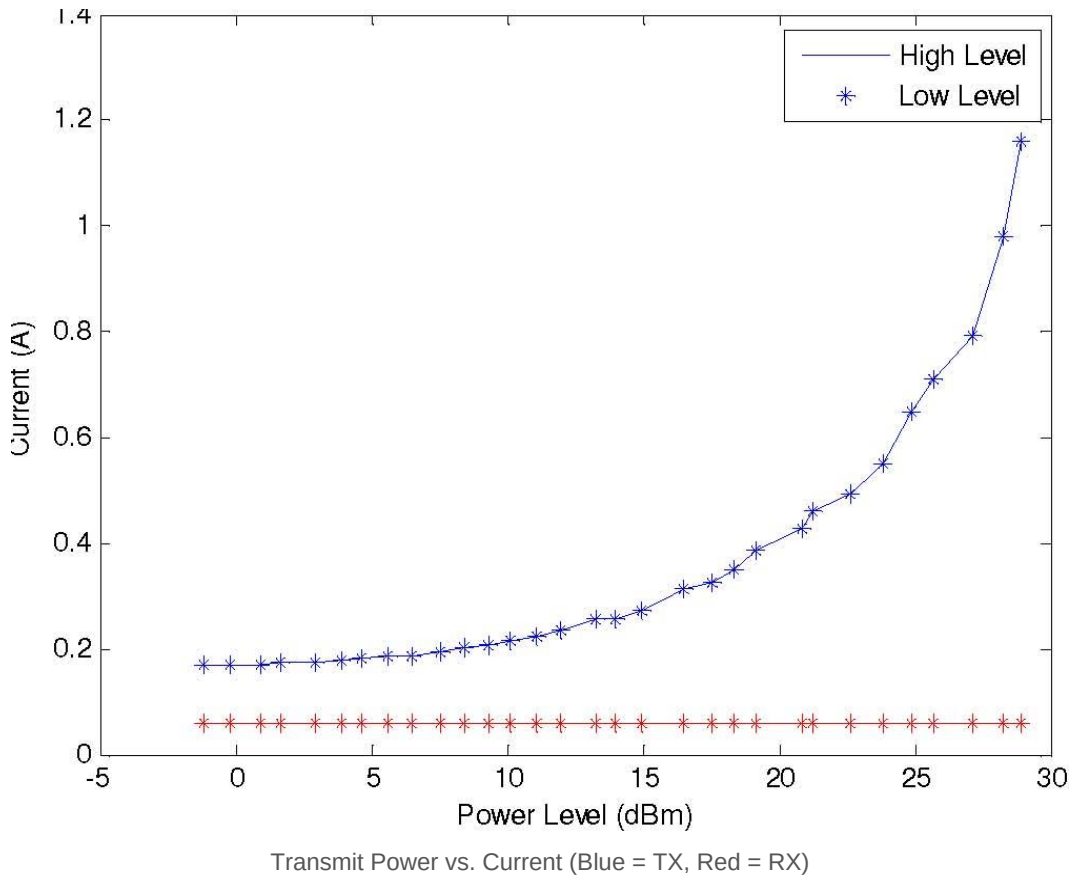


Figure 3.2: The graph below shows the relationship between transmit power and transmit/receive current. Blue line = transmit current, Red line = receive current.



4. Interface Definitions and LED Indicator Meanings



RFD900A Interface and LED Indicators

LED Meanings:

- **ACT (Green) Indicator:** Flashing → Searching / pairing data transmission in progress
Solid ON → Data transmission pairing / connection successful
- **COM (Red) Indicator:** Flashing → Communication in progress
Solid ON → BootLoader mode active

The module has two SMA antenna connectors (labeled 1# and 2#), UART and USB interfaces. USB takes priority when both are connected.

5. Software Introduction

The RFD900A software is an open-source system compatible with RFD900Tools, 3DR Radio Config, SikRadio configuration tools, and the data transmission configuration module integrated in Mission Planner.

RFD900A includes a built-in BootLoader supporting USB online firmware updates.

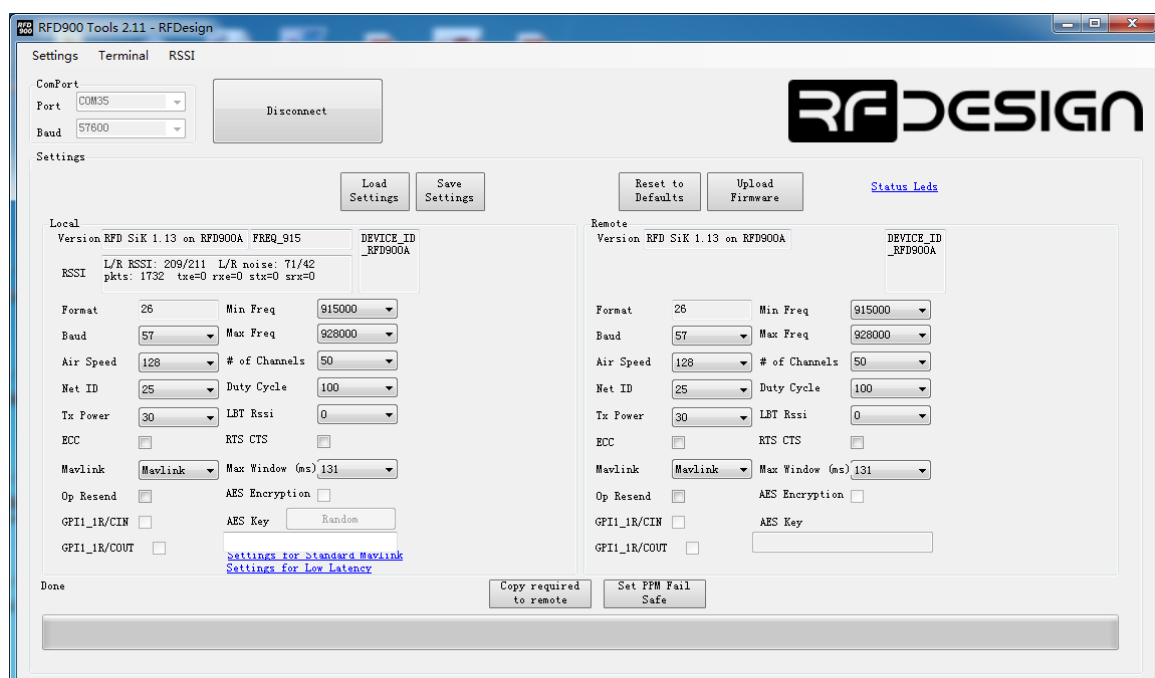
The RFD900A data transmission module offers rich software configuration features, including:

- Frequency hopping range setting
- User-configurable serial communication rate and air communication rate
- Transmit power adjustment
- ECC error correction, optional Mavlink protocol frame communication
- MavLink radio status reporting (local RSSI, remote RSSI, local noise, remote noise)

Software Interface Overview

Main functions available in the configuration software:

- **Load settings:** Load parameters from the module
- **Save settings:** Write current parameters to the module
- **Reset to defaults:** Load factory default parameters
- **Upload firmware:** Upload new firmware
- **Version:** Firmware version (1.X = point-to-point, 2.X = point-to-multipoint)
- **RSSI:** Wireless signal status — L/R RSSI shows local/remote signal strength (higher = better). L/R noise shows local/remote noise levels reflecting interference in the band.
- **Baud:** Serial baud rate adjustment (default 57 = 57600)
- **Min / Max Freq:** Frequency hopping range adjustment
- **Air Speed:** Air communication rate adjustment
- **Net ID:** Network ID for pairing different modules
- **Duty Cycle:** Transmission duty cycle adjustment



RFD900A Configuration Software Interface

TX POWER – Transmit Power Settings

dBm setting values and corresponding power in watts:

dBm	Watts	dBm	Watts	dBm	Watts
0	1.0 mW	10	10 mW	20	100 mW
1	1.3 mW	11	13 mW	21	128 mW
2	1.6 mW	12	16 mW	22	160 mW
3	2.0 mW	13	20 mW	23	200 mW
4	2.5 mW	14	25 mW	24	250 mW
5	3.2 mW	15	32 mW	25	320 mW
6	4.0 mW	16	40 mW	26	400 mW
7	5.0 mW	17	50 mW	27	500 mW
8	6.0 mW	18	64 mW	28	640 mW
9	8.0 mW	19	80 mW	29	800 mW
				30	1000 mW (1W)

6. Antenna Selection

The RFD900 has two SMA antenna interfaces and supports antenna diversity technology. When the module is in receive mode, it automatically detects both antennas and selects the one with the better received signal. If only one antenna is connected, the system automatically selects that interface. Silicon Labs testing shows that antenna diversity can effectively compensate for 6–8 dB of attenuation.

RFD900 primarily uses two antenna diversity techniques:

Spatial Diversity

Spatial diversity requires placing the two antennas a certain distance apart. The recommended separation is more than $0.6 \times$ the wavelength and an odd multiple of $\frac{1}{4}$ wavelength (e.g., $\frac{3}{4}$ or $\frac{5}{4}$). At 915 MHz the wavelength is approximately 32.8 cm, so the two antennas should be separated by at least 25 cm. Greater separation is better if possible.

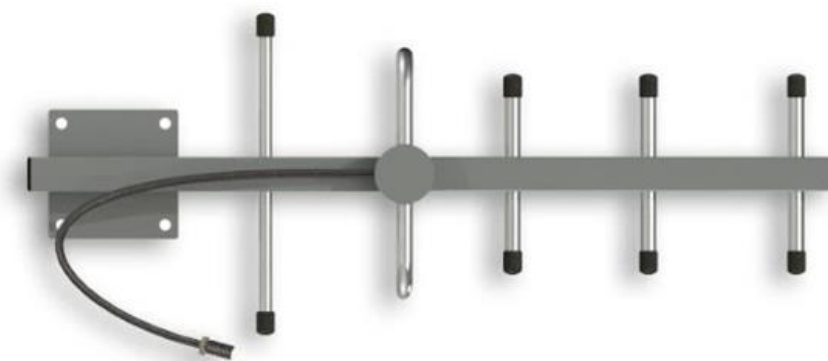
Polarization Diversity

Polarization diversity uses the two polarization characteristics of the wireless signal (one vertical, one horizontal). Use one horizontally polarized antenna and one vertically polarized antenna.

Optional Antenna Solutions

Yagi Antenna

The director elements of a Yagi antenna provide excellent signal superposition and higher gain compared to dipole antennas. It performs particularly well for direction finding and long-distance communication. Due to its size and high directivity, it is best used at the ground station end, ideally mounted on an AAT (Automatic Antenna Tracker) system for optimal results.



Yagi Antenna Example

$\frac{1}{4}$ -Wave Monopole Antenna

Recommended for diversity mounting on the airborne (UAV) end.



$\frac{1}{4}$ -Wave Monopole Antenna

$\frac{1}{2}$ -Wave Dipole Antenna

A commonly used antenna with approximately omnidirectional radiation pattern. Suitable for both airborne and ground station installation. Transmission distance is shorter than a Yagi antenna.



$\frac{1}{2}$ -Wave Dipole Antenna

7. AT Command Set

The RFD900 supports configuration via the Hayes AT command set through a terminal. To enter AT command mode, type **+++** in the terminal. Upon successful entry, the screen returns **OK**. In AT mode you can control the local RFD900 with AT commands or the remote module with RT commands.

To apply settings via AT commands, complete these three steps in order:

1. Use command **ATSn=X** to set parameter register n to value X.
2. Use command **AT&W**; to write all modified values to EEPROM for permanent storage.
3. Use command **ATZ** to restart the RFD900 module and apply changes.

AT Command List

Command	Description
ATI	Display radio version
ATI2	Display board type
ATI3	Display board frequency
ATI4	Display board version
ATI5	Display all user-configurable parameters
ATI6	Display TDM timing report
ATI7	Display RSSI signal report
ATO	Exit AT command mode
ATSn?	Display value of parameter register n
ATSn=X	Set parameter register n to value X
ATZ	Restart module
AT&W	Write current parameters to EEPROM
AT&F	Restore all parameters to factory defaults
AT&T=RSSI	Enable RSSI debug reporting
AT&T=TDM	Enable TDM debug reporting
AT&T	Disable debug reporting

RFD900 Parameter Register List

Reg	Name	Description	Default	Max	Min	Local=Remote?
0	FORMAT	EEPROM version (read-only)	FW dep.	N/A	N/A	No
1	SERIAL_SPEED	Serial baud rate	57	115	2	No
2	AIR_SPEED	Air data rate	64/128	250	2	Yes
3	NET ID	Network ID (must match)	25	499	0	Yes
4	TX POWER	Transmit power (dBm)	20/27	30	0	No
5	ECC	Enable Golay ECC	1	1	0	Yes
6	MAVLINK	Enable Mavlink + reports	1	1	0	No

7	OP_RESEND	Opportunistic Resend	1	1	0	No
8	MIN_FREQ	Minimum frequency (kHz)	915000	927000	902000	Yes
9	MAX_FREQ	Maximum frequency (kHz)	928000	928000	903000	Yes
10	NUM_CHANNELS	Number of channels	50/20	50	5	Yes
11	DUTY_CYCLE	TX duty cycle (%)	100	100	10	No
12	LBT_RSSI	RSSI LBT control (do not modify)	0	1	0	Yes
13	MANCHESTER	Manchester encoding (do not modify)	0	1	0	Yes
14	RTSCTS	RTS/CTS control (do not modify)	0	1	0	No
15	NODEID	Local node ID (multipoint)	2	29	0	N/A
16	NODEDEST	Destination node ID (65535=broadcast)	65535	29	0	No
17	SYNCANY	Allow sync to non-base (set 0 recommended)	0	1	0	No
18	NODECOUNT	Total node count (multipoint)	3	30	2	Yes

Note: Registers 16, 17, and 18 require multipoint (2.X) firmware.

8. About Air Data Rate

Wireless data transmission performance is affected by latency, signal interference, error correction, and other factors. Effective range decreases as distance increases. Therefore, the AIR_SPEED setting should be chosen according to actual requirements.

Example: With good antenna performance, setting AIR_SPEED to 64 kbps typically allows ~40 km range. Lowering AIR_SPEED can increase range further, but may not satisfy required data throughput. For short-range links you can raise AIR_SPEED to increase throughput. Choose a balance between speed and range.

Available AIR_SPEED values: 2, 4, 8, 16, 19, 24, 32, 48, 64, 96, 128, 192, 250 kbps

Recommendation: Set serial baud rate lower than air rate (e.g., air speed 64 kbps → baud rate 57 = 57600 bps).

9. Common Links

RFD900 Firmware Download

<http://rfdesign.com.au/firmware/>

V1.x = Point-to-point firmware | V2.x = Multipoint firmware

RFD900 Modem Tools

<http://rfdesign.com.au/downloads/>

Software Solution Sik V1.x

<https://github.com/RFDesign/SiK>

3DR Radio Configuration Tool

<http://vps.oborne.me/3drradioconfig.zip>

This document is an English translation of the original Chinese RFD900A User Manual. For the most up-to-date information, please refer to the official RFDesign website.